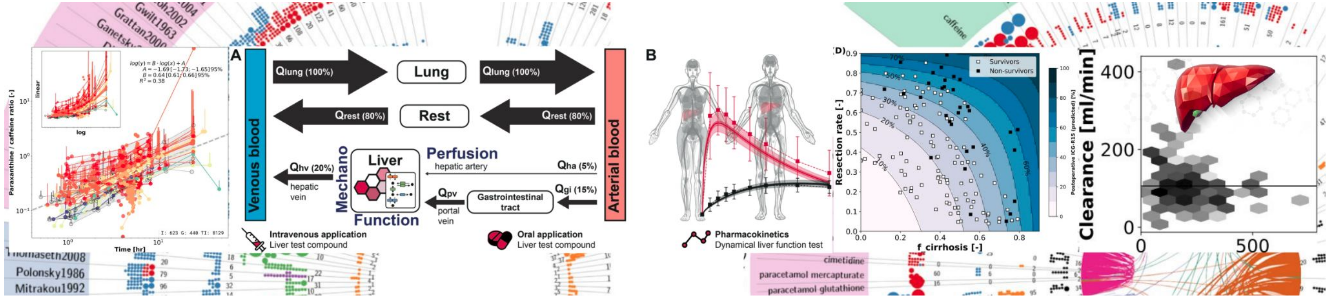


Pharmacokinetics Modeling Course

Reproducible Research & SBML

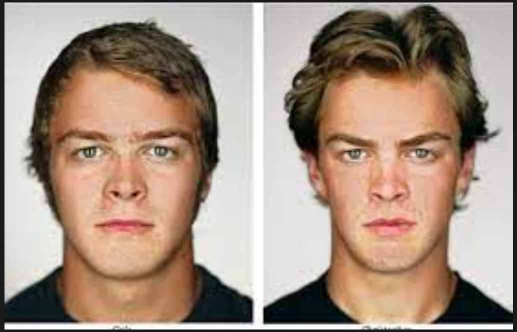
Dr. Matthias König
Humboldt-University Berlin
Systems Medicine of the Liver
koenigmx@hu-berlin.de
https://livermetabolism.com

🎯 Teaching Goals

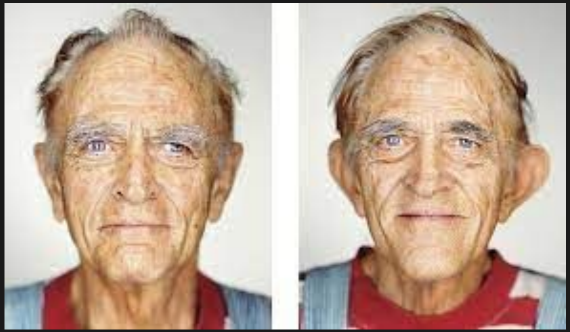
By the end of this section, you should be able to:

1. **Understand the role of reproducibility** in computational modeling.
2. **Introduce the Systems Biology Markup Language (SBML)** as a standard for model representation and exchange.
3. **Gain familiarity with tools** for creating,and simulating SBML models (e.g., sbmlutils, cy3sbml).
4. **Discuss FAIR principles** (Findable, Accessible, Interoperable, Reusable) and their application to model data and metadata.
5. **Highlight the importance of open science practices** for transparent and collaborative research.

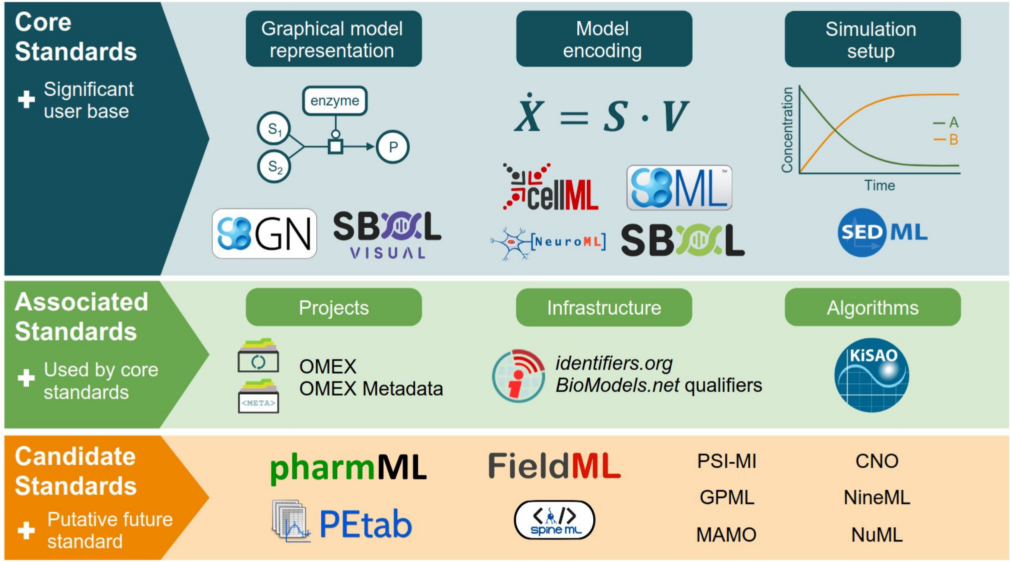




REPRODUCIBILITY



Development of Standards



- Models (SBML)
- Simulationen (SED-ML)
- Visualisation (SGBN)
- Parameter Optimization (PETab)
- Metadata (OMEX)
- COMBINE archive

Specifications of Standards in Systems and Synthetic Biology: Status and Developments in 2022 and the COMBINE meeting 2022. **M. König** et al. J Integr Bioinform. 2023 Mar 29;20(1).

SBML Level 3: an extensible format for the exchange and reuse of biological models. SM Keating, D Waltemath, **M König**, ..., M Hucka, and SBML Community members. Mol Syst Biol. 2020;16(8):e9110. doi:10.15252/msb.20199110

The Systems Biology Markup Language (SBML): Language Specification for Level 3 Version 2 Core. M. Hucka, F. Bergmann, C. Chaouiya, A. Dräger, S. Hoops, S. Keating, **M. König**, N Le Novere, C. Myers, B. Olivier, S. Sahle, J. Schaff, R. Sheriff, L. Smith, D. Waltemath, D. Wilkinson, F. Zhang. J Integr Bioinform. 2019 Jun 20;16(2); 10.1515/jib-2019-0021

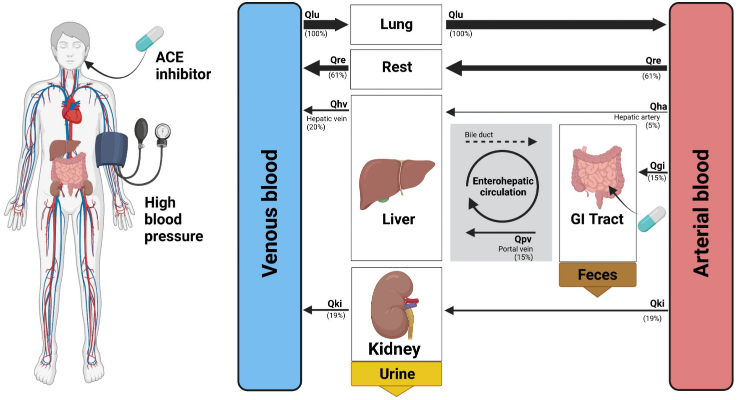
The Distributions Package for SBML Level 3. L Smith, S Moodie, F Bergmann, C Gillespie, S Keating, **M. König**, C Myers, M Swat, D Wilkinson, M Hucka. J Integr Bioinform. 2020 Aug 4; doi: 10.1515/jib-2020-0018

Open modeling and exchange (OMEX) metadata specification version 1.0. Neal ML, Gennari JH, Waltemath D, Nickerson DP, **König M.** J Integr Bioinform. 2020 Jun 25;17(2-3):20200020. doi:10.1515/jib-2020-0020.

Harmonizing semantic annotations for computational models in biology. Neal, M.; **König,** M.; Nickerson, D., ...mWaltemath, D. Brief Bioinform. 2019 Mar 22;20(2):540-550. doi:10.1093/bib/bby087.

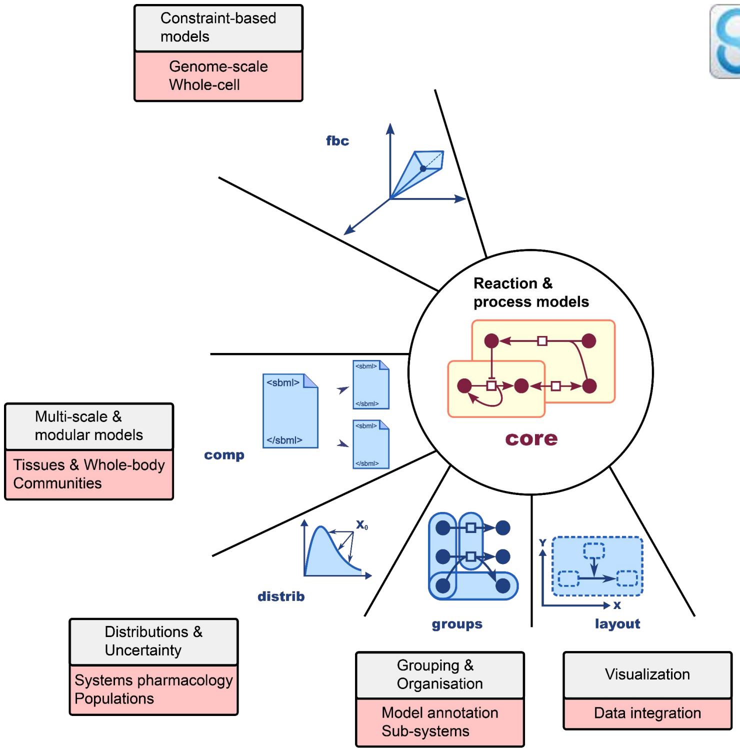
Systems Biology Markup Language (SBML)

- SBML is a software data format for describing models in biology (https://sbml.org)
- It's a little like HTML but for model instead of web pages
- independent of any particular tool
- free and open
- De facto standard



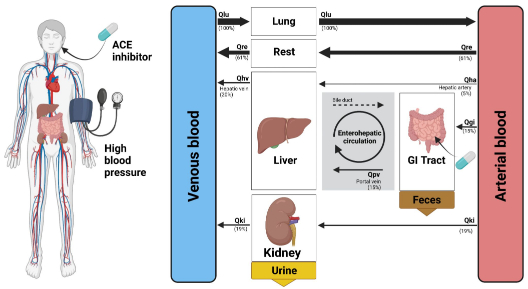
SBML

- **Process-based ODE models**
- Reproducible & exchangeable model encoding (**SBML**)
- Annotations to modelling, biological and medical ontologies (**SBML core**)
- Hierarchical models/multi-scale models
- (**SBML comp**)
- Unit validation, unit checking, unit conversion
- Distributions in models & uncertainty in data and parameters (**SBML distrib**)
- Mass- & charge balance (**SBML fbc**)
- Use wide range of tools (visualization, parameter fitting, simulation, ...)
- http://sbml.org

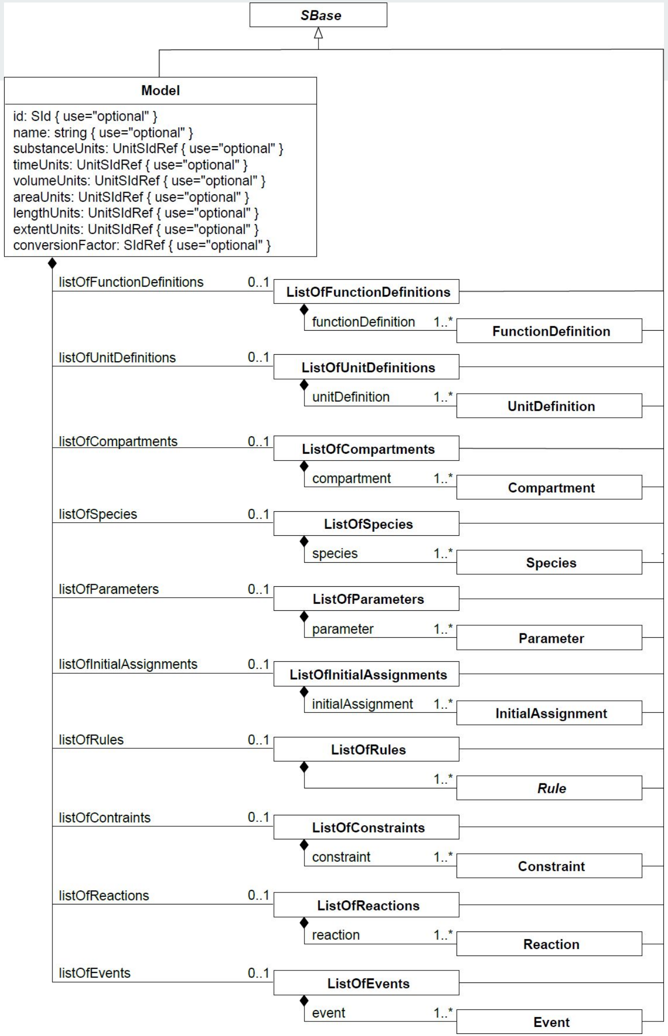


Keating SM, Waltemath D, König M, ... , Hucka M; SBML Level 3 Community members. SBML Level 3: an extensible format for the exchange and reuse of biological models. Mol Syst Biol. 2020 Aug;16(8):e9110. doi: 10.15252/msb.20199110.

SBML core



- Compartments
  - containers/volumes (e.g. liver volume)
- Species
  - molecules in compartments (e.g. `glucose` in plasma)
- Parameters
  - parameters with values (e.g. `cardiac output`)
- AssignmentRules
  - mathematical relationships between other parameters, species, compartments (e.g. `BMI = bodyweight/height`^{2})
- Reactions
  - processes converting species in other species (e.g. enzymatic conversion; e.g. transport via blood flow)



sbmlutils: Python Utilities for Working with SBML

https://github.com/matthiaskoenig/sbmlutils

Packages

- core, fbc, comp, distrib, layout

Features

- model creation, manipulation & merging
- unit support
- model annotation
- interpolations
- file converters (XPP)
- markdown annotations
- OMEX support

```
Parameter(
    "ICGIM_ki_bil",
    0.02,
    unit=U.mM,
    name="Ki bilirubin of icg import",
    sboTerm=SBO.INHIBITORY_CONSTANT,
    notes="""
    bilirubin reference range:
    ```
    ~ 0.1 - 5 [g/dl]
    (10 [mg/l] /584.6623 [g/mole]) ~ 0.0171 mmole/l`
    ```

    setting Ki in reference range
    """,
),
```

Parameter

ICGIM_ki_bil Ki bilirubin of icg import

id	ICGIM_ki_bil
metaID	meta_ICGIM_ki_bil
name	Ki bilirubin of icg import
sbo	SBO:0000261
value	0.02
constant	✔
units	$\frac{mmol}{l}$
derivedUnits	$\frac{mmol}{l}$
cvterms	

BQB_IS sbo SBO:0000261

inhibitory constant

Synonym: Ki

notes

bilirubin reference range:

```
    ~ 0.1 - 5 [g/dl]
(10 [mg/l] /584.6623 [g/mole]) ~ 0.0171 mm
```

setting Ki in reference range

SBML4Humans: Human Readable SBML Report

https://sbml4humans.de

- **interactive SBML report** with navigation between SBML objects
- **web application** (no setup)
- **search** and filter functionality
- **resolve/render metadata**
- **hierarchical models** (SBML comp)
- **distributions and uncertainties** (SBML distrib)
- **flux balance** (SBML fbc)
- **COMBINE archives** (multiple models)
- **URL endpoint** for integration in tools/ workflows/ webpages/ presentations

https://sbml4humans.de/model_url?url=https://www.ebi.ac.uk/biomodels/model/download/BIOMD0000000001.2?filename=BIOMD0000000001_url.xml

Parameter

HCT hematocrit

id	HCT
metaID	meta_HCT
name	hematocrit
sbo	SBO:0000002
value	0.51
constant	✓
units	–
derivedUnits	–
cvterms	

BQB_IS sbo SBO:0000002

quantitative systems description parameter
A numerical value that defines certain characteristics of systems or system functions. It may be part of a calculation, but its value is not determined by the form of the equation itself, and may be arbitrarily assigned.

BQB_IS ncit C64796

Hematocrit Measurement
A measure of the volume of red blood cells expressed as a percentage of the total blood volume. Normal in males is 43-49%, in females 37-43%.
Synonyms
- HCT
- Packed Cell Volume
- Hematocrit
- Erythrocyte Volume Fraction
- PCV
- Hematocrit Measurement
- EVF

BQB_IS omit 0007571

Hematocrit

BQB_IS efo 0004348

hematocrit
The volume of packed RED BLOOD CELLS in a blood specimen. The volume is measured by centrifugation in a tube with graduated markings, or with automated blood cell counters. It is an indicator of erythrocyte status in disease.

Parameter (1)

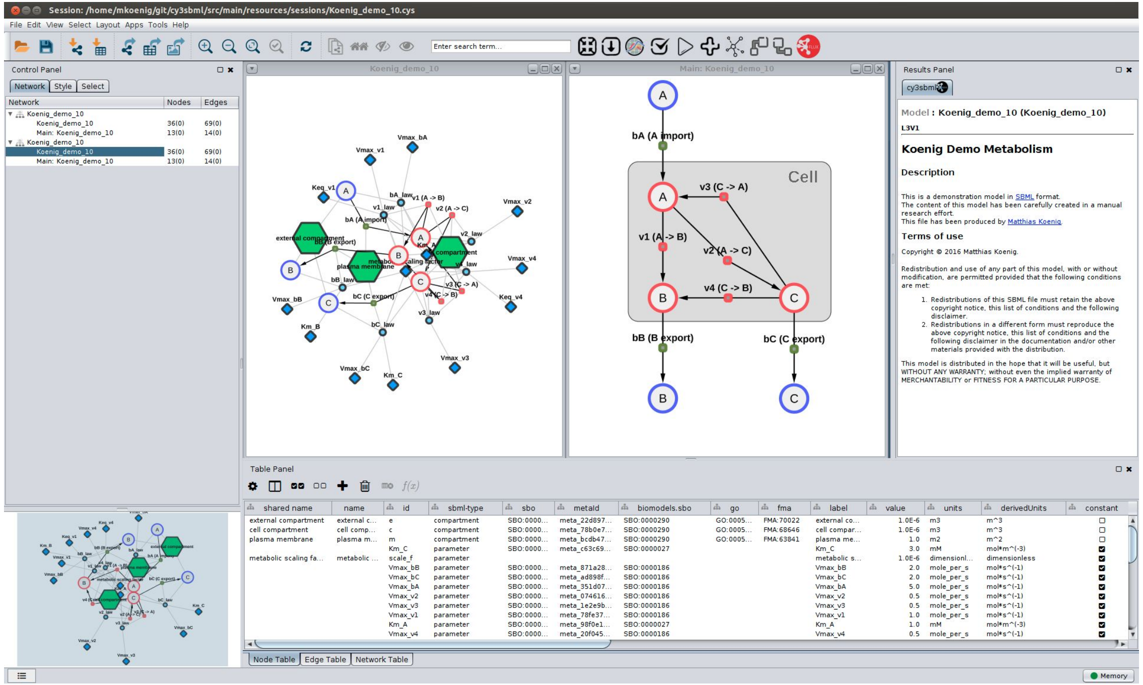
id	name	constant	value	units	derivedUnits	assignment
HCT	hematocrit	✓	0.51	–	–	

AssignmentRule (8)

id	name	variable	math	derivedUnits
Vve		Vve	$(1-HCT)\cdot(BW\cdot FVve - \frac{FVve}{FVar+FVve+FVpo+FVhv}\cdot BW\cdot Fblood\cdot(1-(FVar+FVve+FVpo+FVhv)))$	l
Var		Var	$(1-HCT)\cdot(BW\cdot FVar - \frac{FVar}{FVar+FVve+FVpo+FVhv}\cdot BW\cdot Fblood\cdot(1-(FVar+FVve+FVpo+FVhv)))$	l
Vpo		Vpo	$(1-HCT)\cdot(BW\cdot FVpo - \frac{FVpo}{FVar+FVve+FVpo+FVhv}\cdot BW\cdot Fblood\cdot(1-(FVar+FVve+FVpo+FVhv)))$	l
Vhv		Vhv	$(1-HCT)\cdot(BW\cdot FVhv - \frac{FVhv}{FVar+FVve+FVpo+FVhv}\cdot BW\cdot Fblood\cdot(1-(FVar+FVve+FVpo+FVhv)))$	l
Vre_plasma		Vre_plasma	$Vre\cdot Fblood\cdot(1-HCT)$	l
Vgi_plasma		Vgi_plasma	$Vgi\cdot Fblood\cdot(1-HCT)$	l
Vli_plasma		Vli_plasma	$Vli\cdot Fblood\cdot(1-HCT)$	l
Vlu_plasma		Vlu_plasma	$Vlu\cdot Fblood\cdot(1-HCT)$	l

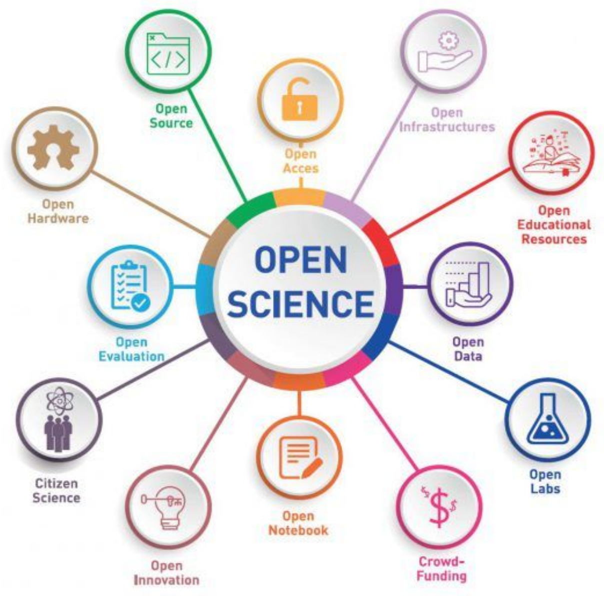
CySBML: Cytoscape app for visualizing SBML

- **kinetic & reaction-species view**
- **subgraphs & filtering**
- **annotation support**
- **works for large scale networks** (genome-scale)
- **sbmlutils integration** (py4cytoscape)



https://github.com/matthiaskoenig/cy3sbml

Open und FAIR Science



FE Kohrs, … , **M König**, …, TL Weissgerber. **Eleven Strategies for Making Reproducible Research and Open Science Training the Norm at Research Institutions.** eLife (2023)

Ten Simple Rules for FAIR Sharing of Experimental and Clinical Data with the Modeling Community. **König M,** Grzegorzewski J, Golebiewski M., Hermjakob H, Hucka M, Olivier B, Keating SM, Nickerson D, Schreiber F, Sheriff R, Waltemath D. Preprints 2021, 2021080303, doi: 10.20944/preprints202108.0303.v2.

FAIR Sharing of Reproducible Models of Epidemic and Pandemic Forecast Ramachandran, K.*; **König, M.***; Scharm, M.; Nguyen, T.V.N.; Hermjakob, H.; Waltemath, D.; Malik Sheriff, R.S. (* equal contribution). Preprints 2022, 2022060137 (doi: 10.20944/preprints202206.0137.v1).

Open Data in Repositories

- BioModels DB, GitHub, Zenodo
- PK-DB

Open/FAIR models

- CC-BY, Zenodo, GitHub, FAIR Indicators

Open Science Initiatives

- Strategy development (Open Science Training at Universities)
- Open Science Ambassador (2024-2025)